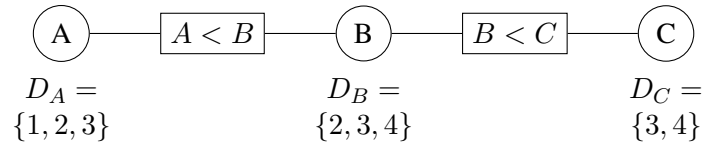


Exercise 1 :



- Which arcs in the above constraint network are arc consistent?
- How can the whole network be made arc consistent, i.e., which changes should be made to the domains of the variables making the network arc-consistent without eliminating potential solutions?

Exercise 2 :

Consider the following constraint network $\gamma = (V, D, C)$:

- Variables: $V = \{a, b, c, d\}$.
- Domains: For all $v \in V$: $D_v = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.
- Constraints: $2|a - c| > 3$; $b^2 - 3d < 9$; $b + 3 < c$.

Run the Generalized Arc Consistency(γ) algorithm, as specified in the lecture. Precisely, for each iteration of the while-loop, give the content of To-do-arcs at the start of the iteration, give the pair (x, c) removed from To-do-arcs, give the domain D_x of x after the call to $\text{Revise}(\gamma, x, c)$, and give the pairs (y, c') added into M .

Note: Initialize To-do-arcs as a lexicographically ordered list (i.e., (a, b) would be before (a, c) , both before (b, a) etc., if any of those exist). Furthermore, use to-do-arcs as a FIFO queue, i.e., always remove the element at the front and add new elements at the back.

Exercise 3 :

Solve exercise 4.14 in PM about variable elimination. (<https://artint.info/3e/html/ArtInt3e.Ch4.S12.html>)

Exercise 4 :

Use Variable Elimination to solve the following CSP given by extensional constraints on Boolean variables A, B, C :

A	B	A	C	B	C
t	f	t	f	t	f
t	t	f	t	f	t
f	t				

Exercise 5 :

Consider the cryptarithmic puzzle

$$\begin{array}{rcccc}
 & T & W & O & \\
 + & T & W & O & \\
 \hline
 F & O & U & R &
 \end{array}$$

Each letter in a cryptarithmic problem represents a different digit; note that F cannot be 0.

- Construct a constraint network representation for the puzzle. *Hint:* For each of the four columns in the table representation above we have a constraint. E.g. for the first column we have the constraint

$$O + O = R + 10 \cdot C_1,$$

where C_1 is an auxiliary variable representing what is carried over in the 10 column. Use similar auxiliary variables, say C_2 and C_3 , for encoding what is carried over to the 100 and 1000 column, respectively.

- Perform a BacktrackingWithInference to find a solution to the puzzle. *Hint:* The ordering of the variables has a big impact on the solution size, so a good strategy would be to choose the variable with the smallest number of possible values. E.g. consider starting with $C_3 = 1$.

Exercise 6 :

Consider the following mini-Sudoku:

		4	
2			
		1	
3			

The empty fields have to be filled with numbers 1,2,3, or 4, such that each row, each column, and each of the 2×2 sub-squares contain each of these numbers exactly once.

- Formalize this problem as a constraint satisfaction problem: define an appropriate set of variables, and a set of constraints that define the solutions of the sudoku.
 - Hint 1: instead of using constraints as on the lecture slide, define a smaller set of constraints, each constraint expressing the full condition for one row, one column, or one sub-square. We have not discussed off-the-shelf formal languages for expressing such constraints, so you should try to express them as formally as possible. But don't overthink it, as long as the constraints are not ambiguous, it is fine).
 - Hint 2. To define variables, you can simply label the empty fields with variables like this:

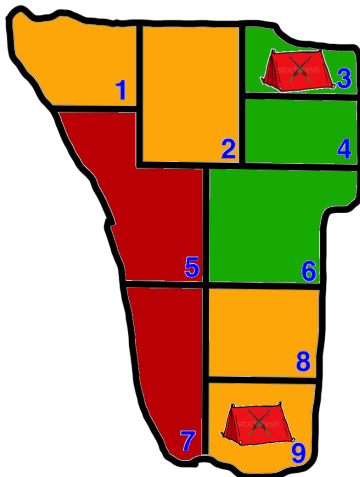
A1	A2	4	A4
2	B2	B3	B4
C1	C2	1	C4
3	D2	D3	D4

- b. Draw the constraint network for this problem. If this gets too large, draw only the part of the network that is sufficient to answer the next question.
- c. Apply the generalized arc consistency algorithm to show that the top left square must contain a 1. Does the operation of the algorithm resemble how you would come to that conclusion yourself?

Exercise 7 :

(Extra exercise for modelling CSPs)

You are a ranger and own a piece of land, depicted below. You want to make sure that all of your animals survive. This is only possible if no animals that kill each other live in the same or neighboring regions. You own rhinos (R), elephants (E), cheetahs (C), lions (L), jackals (J), springbok (S), zebras (Z), hippos (H) and giraffes (G). Your piece of land contains three different climatic regions: bush (green), desert (red), savannah (orange). Some animals need specific climatic environments. Unfortunately there are also two camps of poachers on the land, who are selling rhino horn and elephant tusk. Poachers treacherously shoot animals in their own and neighboring regions.



- Poachers kill rhinos and elephants
- Jackals eat springbok
- Cheetahs eat springbok and zebras
- Lions eat springbok, zebras and giraffes, and sometimes even kill elephants

- Jackals and cheetahs can live in the desert and in the savannah
- Elephants can only live in the bush
- Lions and zebras can live in the bush and in the savannah
- Springbok and hippos only live in the savannah regions

Formulate the problem as a constraint network. Use the set of variables $V = \{v_1, \dots, v_9\}$, where variable v_i models which animal can live in region i .

- (i) Specify the domain D_{v_i} of each variable $v_i \in V$. That is, for each region type give the set of animals that can live in that region, and state which variables have this domain.
- (ii) Specify the binary constraints $C_{v_i v_j} \subseteq D_{v_i} \times D_{v_j}$ for all pairs of variables. That is, for each combination of regions give the set of animal-pairs that the region pair can inhabit, and state which variable pairs have this constraint.
- (iii) Specify the unary constraints $C_{v_i} \subseteq D_{v_i}$ for all relevant variables. That is, for each region affected by the poachers, give the set of animals that can inhabit that region.

Hint: You can specifying constraints with those pairs that are not contained:

$$C_{v_i, v_j} = D_{v_i} \times D_{v_j} \setminus \{(d_{v_i, k}, d_{v_j, l}), \dots\}$$

Exercise 8 :

(Extra exercise to practice arc consistency (a bit long))

On your piece of land you additionally own a lodge for tourists and a cat sanctuary. You want to show your tourists all your nice animals and everything the region has to offer. You have 5 different time slots: 1, 2, 3, 4, 5, and want to fit all the tasks (see below) in, such that all of the constraints (see below) are satisfied.

Tasks:

- (GD) go for game drive with tourists
- (DW) go to the desert with tourists
- (HF) make helicopter flight with tourists
- (HB) have braai with tourists
- (PB) prepare braai
- (PC) pet cheetahs
- (FL) feed lions
- (FC) feed cheetahs
- (FZ) feed zebra

Constraints:

- Cheetahs need to be fed immediately before you can pet them ($FC = PC - 1$)
- Lions and cheetahs have to be fed at the same time, otherwise they will get jealous ($FC = FL$)
- The desert walk has to be (at least) two time slots before the game drive, otherwise your tourists will get to exhausted ($DW \leq GD - 2$)
- In order to have braai with your tourists you first have to exhaust them on the game drive and on the desert walk, so they can enjoy their beer ($GD < HB$; $DW < HB$)
- You have to prepare braai before having the braai with your tourists ($PB < HB$)
- You can only find lions on the game drive if they have been fed two time slots earlier ($GD = FL + 2$)
- In order to give your tourists an impression of the area first, the heli flight is done just before the desert walk ($HF = DW - 1$)
- In order to save time you want to feed the zebra on the game drive ($FZ = GD$)
- Variables: $V = \{DW, FC, FL, FZ, GD, HB, HF, PC, PB\}$
- Domains: For all $v \in V$: $D_v = \{1, 2, 3, 4, 5\}$

- (i) Draw the constraint graph of γ . Label each edge with the corresponding constraint.
 - (ii) Run Generalized Arc consistency. Initialize To-do-arcs with an alphabetical list. For each step: Give the selected pair, the updated domain, and the pairs that are added to To-do-arcs (give all pairs that would be added and mark those that are not already contained). You do not have to give the updated To-do-arcs list in each iteration.
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